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Systems for photocuring liquid resin with reduced heat generation

Abstract

In a vat polymerization printer, a beam scanner scans a light beam across a mask and into a tank containing a photo-curable resin. The mask has pixels configurable to be individually transparent or opaque to portions of the light beam, which has a diameter greater than a cross-sectional dimension of the pixels of the mask. During an exposure time duration, a first subset of the pixels are controlled to be transparent at locations corresponding to the cross section of a three-dimensional object to be printed, while a second subset of the pixels are controlled to be opaque at locations not corresponding to the cross section of the three-dimensional object. The beam scanner is controlled to scan the light beam across the mask such that the light beam is always incident on at least one of the pixels of the mask that are controlled to be transparent.

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Background/Summary

RELATED APPLICATIONS (1) This application is a Divisional application of U.S. application Ser. No. 17/649,050, filed on 26 Jan. 2022, which is a non-provisional patent application of and claims priority to U.S. Provisional Application No. 63/200,258, filed 24 Feb. 2021.

FIELD OF THE INVENTION

(1) The present invention relates to the printing of three-dimensional objects by photo-curing a liquid resin, and more particularly relates to reducing heat imparted into the liquid resin by a light source.

BACKGROUND

(2) One obstacle encountered in the three-dimensional printing of objects that involves the curing of photo-curable liquid resin is the heating of the liquid resin. Not only is the curing of photo-curable liquid resin an exothermic reaction (which locally heats regions of the photo-curable liquid resin where the curing takes place), but the irradiation of a mask by a light source, typically an ultra-violet (UV) light source, also causes heating of the mask. As the mask is located in close proximity to the liquid resin, any heating of the mask also leads to the further heating of the photo-

curable liquid resin.

(3) If the liquid resin temperature exceeds a critical temperature, portions of the resin may start to cure even in the absence of UV light, leading to defects in the printed objects. In prior approaches to prevent the liquid resin temperature from exceeding this critical temperature, the printing process may be periodically halted to allow the photo-curable liquid resin to cool, with the consequence of reducing the throughput of the printing process. Also in prior approaches, a resin circulatory system may be employed to cool the heated resin. While heat removal via a resin circulatory system may effectively achieve the desired effect of controlling the liquid resin temperature, approaches described herein control the temperature of the liquid resin through other or additional means.

SUMMARY OF THE INVENTION

(4) In one embodiment of the present invention, a vat polymerization printer includes a tank configured for containing a photo-curable liquid resin, a light source configured to emit a light beam, and a mask positioned between the light source and the tank and having pixels configurable to be individually transparent or opaque to portions of the light beam. Preferably, a diameter of a cross section of the light beam is greater than a cross-sectional dimension of each of the respective pixels. A beam scanner is configured to scan the light beam across the mask, and a processor operating under stored processor-executable instructions controls the vat polymerization printer to print a cross section of a three-dimensional object by: controlling, during an exposure time duration, a first subset of the pixels to be transparent at locations corresponding to the cross section of the three-dimensional object, and a second subset of the pixels to be opaque at locations not corresponding to the cross section of the three-dimensional object; and controlling, during the exposure time duration, the beam scanner to scan the light beam across the mask such that the light beam is always incident on at least one of the pixels of the mask that are controlled to be transparent.

(5) In various embodiments, the diameter of the cross section of the light beam may be at least ten times or at least a hundred times the cross-sectional dimension of each of the respective pixels of the mask. Further, the light source may include a laser source configured to emit a laser beam; and a beam expander configured to generate the light beam from the laser beam, wherein the diameter of the cross section of the light beam is greater than a diameter of a cross section of the laser beam.

(6) In various embodiments, the processor-executable instructions may further cause the processor to determine a scan path for the light beam based on respective locations of the pixels that are controlled to be transparent during the exposure time duration. Also, during the exposure time duration, the processor-executable instructions may further cause the processor to turn off the light source while the beam scanner repositions the light beam between a first region of the mask that includes at least some pixels that are controlled to be transparent to a third region of the mask that includes at least some pixels that are controlled to be transparent, the third region of the mask being separate from the first region of the mask by a second region of the mask that includes only pixels that are controlled to be opaque. And, in still further embodiments, the processor-executable instructions may further cause the processor to control a blocking element to block the light beam while the beam scanner repositions the light beam from a first region of the mask that includes at least some pixels that are controlled to be transparent to the third region of the mask. In the various embodiments, the pixels may be electrically modulated liquid crystal pixel elements.

(7) In various embodiments, the vat polymerization printer may further include a transparent backing member disposed between the mask and a flexible membrane. Additionally, an extraction plate may be disposed within the tank, and during printing the three-dimensional object formed from cured portions of the photo-curing liquid resin is affixed to the extraction plate. A height adjuster may be configured to control a vertical position of the extraction plate above the mask.

(8) Other embodiments of the invention provide a vat polymerization printer that includes a tank configured for containing a photo-curable liquid resin, a light source configured to emit a light beam, and a mask having pixels configurable to be individually transparent or opaque to portions of

the light beam. A diameter of a cross section of the light beam is greater than a cross-sectional dimension of each of the respective pixels and a beam scanner is configured to scan the light beam across the mask. A processor of a controller executes instructions to control the vat polymerization printer to print a cross section of a three-dimensional object by controlling, during an exposure time duration, a first subset of the pixels to be transparent at locations corresponding to the cross section of the three-dimensional object, and a second subset of the pixels to be opaque at locations not corresponding to the cross section of the three-dimensional object; and controlling, during the exposure time duration, the beam scanner to scan the light beam across at least one region of the mask having pixels that are controlled to be transparent, wherein at most ten percent of the pixels that are controlled to be opaque are scanned by the light beam during the printing of the cross section of the three-dimensional object.

(9) In various embodiments, the processor of the controller may further execute instructions to control the beam scanner to repeatedly scan the light beam across a first region of the mask that includes at least some pixels that are controlled to be transparent, followed by controlling the beam scanner to scan the light beam along a beam path within a second region that separates the first region from a third region, the second region including only pixels that are controlled to be opaque, and the third region including at least some pixels that are controlled to be transparent, and the beam path within the second region being a shortest path that connects a beam path in the first region and a beam path in the third region, and followed by controlling the beam scanner to repeatedly scan the light beam across the third region of the mask. Repeatedly scanning the light beam across the first region of the mask comprises at least one of a raster scan or a back and forth scan of the first region of the mask, and repeatedly scanning the light beam across the third region of the mask comprises at least one of a raster scan or a back and forth scan of the third region of the mask.

(10) Another embodiment of the invention provides for printing a cross section of a three-dimensional object in a photocuring region of a vat polymerization printer that includes (i) a tank configured for containing a photo-curable liquid resin, (ii) a flexible membrane defining a bottom boundary of the photocuring region, (iii) a light source configured to emit a light beam, (iv) a beam scanner configured to scan the light beam, and (v) a mask disposed between the beam scanner and the flexible membrane and having pixels configurable to be individually transparent or opaque to portions of the light beam, wherein a diameter of a cross section of the light beam is greater than a cross-sectional dimension of each of the respective pixels. According to the printing process, during an exposure time duration a first subset of the pixels are controlled to be transparent at locations corresponding to the cross section of the three-dimensional object, a second subset of the pixels are controlled to be opaque at locations not corresponding to the cross section of the three-dimensional object, and the light beam is scanned across at least one region of the mask having at least some pixels that are controlled to be transparent and into the photocuring region, wherein at most ten percent of the pixels that are controlled to be opaque are scanned by the light beam during the printing of the cross section of the three-dimensional object.

(11) In this printing process, during the exposure time duration, and as a result of the control of the first and second subset of the pixels, a first region of the mask includes at least some pixels that are controlled to be transparent, a second region of the mask includes only pixels that are controlled to be opaque, and a third region of the mask includes at least some pixels that are controlled to be transparent, and the scanning of the light beam comprises repeatedly scanning the light beam across the first region of the mask and into the photocuring region through pixels in the first region that are controlled to be transparent, followed by scanning the light beam along a shortest path, within the second region, that connects a beam path in the first region and a beam path in the third region, and followed by repeatedly scanning the light beam across the third region of the mask and into the photocuring region through pixels in the third region that are controlled to be transparent. Repeatedly scanning the light beam across the first region of the mask comprises at least one of a

raster scan or a back and forth scan of the first region of the mask, and wherein repeatedly scanning the light beam across the third region of the mask comprises at least one of a raster scan or a back and forth scan of the third region of the mask.

(12) Alternatively, or in addition, during the exposure time duration, and as a result of the control of first and second subset of the pixels, a first region of the mask includes at least some pixels that are controlled to be transparent, a second region of the mask includes only pixels that are controlled to be opaque, and a third region of the mask includes at least some pixels that are controlled to be transparent, and the scanning of the light beam comprises repeatedly scanning the light beam across the first region of the mask and into the photocuring region through pixels of the first region that are controlled to be transparent, repositioning the light beam from the first region of the mask to the third region of the mask without scanning the second region of the mask, and repeatedly scanning the light beam across the third region of the mask and into the photocuring region through pixels of the third region that are controlled to be transparent. Repeatedly scanning the light beam across the first region of the mask comprises at least one of a raster scan or a back and forth scan of the first region of the mask, and wherein repeatedly scanning the light beam across the third region of the mask comprises at least one of a raster scan or a back and forth scan of the third region of the mask.

(13) During the exposure time duration of the printing process, a total number of pixels in the first subset of the pixels may be less than a total number of pixels in the second subset of the pixels.

(14) Still another embodiment of the invention provides for printing a cross section of a three-dimensional object in a photocuring region of a vat polymerization printer that includes (i) a tank configured for containing a photo-curable liquid resin, (ii) a flexible membrane defining a bottom boundary of the photocuring region, (iii) a light source configured to emit a light beam, (iv) a beam scanner configured to scan the light beam, and (v) a mask disposed between the beam scanner and the flexible membrane and having pixels configurable to be individually transparent or opaque to portions of the light beam, wherein a diameter of a cross section of the light beam is greater than a cross-sectional dimension of each of the respective pixels. The process includes controlling, during an exposure time duration, a first subset of the pixels to be transparent at locations corresponding to the cross section of the three-dimensional object, and a second subset of the pixels to be opaque at locations not corresponding to the cross section of the three-dimensional object; and scanning, during the exposure time duration, the light beam across at least one region of the mask having at least some pixels that are controlled to be transparent and into the photocuring region, wherein the scanning compensates for a non-uniformity in a light transmission across respective pixels in the at least one region of the mask by at least one of: (i) varying a light intensity of the light beam while the light beam is scanned over the at least one region, (ii) varying a scan speed of the light beam while the light beam is scanned over the at least one region, or (iii) varying a number of times the light beam is repeatedly scanned over the at least one region.

(15) These and further embodiments of the present invention are described more fully below.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention is now described, by way of example and without limiting the scope of the invention, with reference to the accompanying drawings which illustrate embodiments of it, in which:

(2) FIG. 1 depicts a cross-section of a three-dimensional (3D) printing system, in accordance with one embodiment of the invention.

(3) FIG. 2 depicts components of a light source, in accordance with one embodiment of the invention.

- (4) FIG. 3A depicts transparent and opaque regions of a mask, in accordance with one embodiment of the invention.
- (5) FIG. 3B depicts a magnified view of a portion of the mask depicted in FIG. 3A, in which individual transparent and opaque pixels are visible, in accordance with one embodiment of the invention.
- (6) FIG. 4A depicts a beam spot of a light beam illuminating a portion of the mask depicted in FIG. 3A, in accordance with one embodiment of the invention.
- (7) FIG. 4B depicts a magnified view of a portion of the mask and beam spot depicted in FIG. 3A, in which individual transparent and opaque pixels are visible, in accordance with one embodiment of the invention.
- (8) FIG. 5 depicts the beam path of a light beam performing a raster scan of a restricted region of the mask (depicted in FIG. 3A) with transparent pixels, in accordance with one embodiment of the invention.
- (9) FIG. 6 depicts transparent and opaque regions of a mask, in accordance with one embodiment of the invention.
- (10) FIG. 7 depicts a back and forth scan of a restricted region of the mask (depicted in FIG. 6) with transparent pixels, in accordance with one embodiment of the invention.
- (11) FIG. 8 depicts transparent and opaque regions of a mask, in accordance with one embodiment of the invention.
- (12) FIG. 9 depicts the beam path of a light beam scanning the mask of FIG. 8, in which the light beam repeatedly scans a first region and a second region with transparent pixels, in accordance with one embodiment of the invention.
- (13) FIG. 10 depicts the beam path of a light beam scanning the mask of FIG. 8, in which the light beam repeatedly scans a first region and a second region with transparent pixels, and further scans a shortest path that connects a beam path in the first region with a beam path in the second region, in accordance with one embodiment of the invention.
- (14) FIG. 11 depicts a 3D printing system with two light beams, each configured to scan a mask of the vat polymerization printer, in accordance with one embodiment of the invention.
- (15) FIG. 12A depicts a flow chart of a method to print a cross section of a three-dimensional object with reduced heat generation, in accordance with one embodiment of the invention.
- (16) FIG. 12B depicts a flow chart of another method to print a cross section of a three-dimensional object with reduced heat generation, in accordance with one embodiment of the invention.
- (17) FIG. 13A depicts a flow chart of a method to scan a light beam across a surface of a mask of a 3D printing system, in accordance with one embodiment of the invention.
- (18) FIG. 13B depicts a flow chart of another method to scan a light beam across a surface of a mask of a 3D printing system, in accordance with one embodiment of the invention.
- (19) FIG. 14 depicts a flow chart of a method to print a cross section of a three-dimensional object with reduced heat generation, and to further compensate, during the printing, for the non-uniformity in the light transmissivity across respective pixels of a mask of a 3D printing system, in accordance with one embodiment of the invention.
- (20) FIG. 15 depicts components of a computer system in which computer readable instructions instantiating the methods of the present invention may be stored and executed.

DETAILED DESCRIPTION

(21) In the following detailed description of embodiments of the invention, reference is made to the accompanying drawings that form a part hereof, and in which are shown by way of illustration specific embodiments in which the invention may be practiced. It is understood that other embodiments may be utilized and structural changes may be made without departing from the scope of the present invention. Descriptions associated with any one of the figures may be applied to different figures containing like or similar components/steps. While the sequence diagrams each present a series of steps in a certain order, the order of some of the steps may be changed.

(22) In one embodiment of the invention, the need to cool the liquid resin is reduced by reducing the degree to which the liquid resin is heated. While the heating of the liquid resin due to the exothermic reaction that takes place during the curing of resin cannot be avoided, the heating of the mask can be reduced by selectively illuminating only regions of the mask with transparent pixels and/or minimizing the illumination of the regions of the mask with opaque pixels. These and other embodiments of the invention are more fully described in association with the drawings below.

(23) FIG. 1 depicts a cross-section of a three-dimensional (3D) printing system **100** (also called a vat polymerization printer), in which electromagnetic radiation (e.g., ultra-violet light) is used to cure photo-curable liquid resin **18** in order to fabricate object **22** (e.g., a 3D object). Object **22** may be fabricated layer by layer; that is, a new layer of object **22** may be formed by photo-curing a layer **34** of liquid resin **18** adjacent to the bottom surface of object **22**, the object may be raised by extractor plate **20**, allowing a new layer of photo-curing liquid resin **18** to be drawn under the newly formed layer; and the process repeated to form additional layers.

(24) The 3D printing system **100** includes tank **10** for containing the photo-curable liquid resin **18**. The bottom of tank **10** includes a bottom opening **12** to allow electromagnetic radiation (e.g., filtered light beam **32**) from light source **24** to enter into tank **10**. An optional radiation-transparent backing member **16** (e.g., borosilicate glass or a toughened glass such as an alkali-aluminosilicate glass of approximately 100 μm thickness) may be used to seal the tank opening **12** (i.e., to prevent the photo-curing liquid polymer **18** from leaking out of tank **10**), while at the same time, allowing electromagnetic radiation to enter into tank **10** in order to cure the liquid polymer.

(25) One challenge faced by 3D printing systems of the present kind is that in addition to adhering to the object **22**, newly formed layers tend to adhere to the bottom of tank. Consequently, when the extraction plate **20** to which the object is attached is raised by height adjustor **39**, the newly formed layer could tear and/or become dissociated from the object **22**. To address this issue, a flexible membrane **14** may be disposed adjacent to backing member **16** (if present) or may form the bottom of the tank (if no backing member is used). Flexible membrane **14** may be formed of silicone or another material, and optionally, coated with a non-stick material such as polytetrafluoroethylene (PTFE) to reduce the likelihood for the newly formed layer to adhere to the bottom of tank **10**. The flexible membrane **14** is transparent (or nearly so) to the wavelength of radiation emitted by the light source **24** so as to allow that radiation to enter into tank **10** in order to cure the liquid polymer **18**.

(26) A mask **30** may be disposed to spatially filter the radiation that is incident on layer **34**, so that specific regions of the liquid resin **18**, that correspond to the cross section of the object **22** being printed, are cured. Mask **30** may be a transmissive spatial light modulator, such as a liquid crystal display (LCD) with a two-dimensional array of addressable pixels. As will be more clearly described below, certain ones of the pixels of the mask may be controlled to be transparent, while others may be controlled to be opaque. Transparent pixels allow radiation to pass through the mask **30** at certain spatial locations of mask **30** and into tank **10**, consequently curing corresponding portions (voxels) of the liquid resin **18**, while opaque pixels prevent radiation from passing through certain spatial locations of mask **30**, thereby avoiding curing of corresponding portions (voxels) of the liquid resin **18**.

(27) A beam scanner **26** may scan light beam **28** across mask **30**. As will be described in more detail below, beam scanner **26** may be controlled by controller **36** to selectively scan light beam **28** across regions of mask **30** with transparent pixels, while substantially avoiding regions of mask **30** with only opaque pixels. Beam scanner **26** may be an x-y scanner, such as a galvo scanner (also known as a galvanometer scanner). In a preferred embodiment (although not depicted in FIG. 1), the distance separating beam scanner **26** from mask **30** is substantially greater than the lateral dimensions of mask **30** so that light beam **28** is incident upon mask **30** at substantially 90° regardless of whether light beam **28** is scanning peripheral regions of mask **30** or central regions of mask **30**. Such placement of the beam scanner **26** relative to the mask **30**, along with a minimal

separation between mask **30** and resin layer **34** decrease the effects of diffraction as light passes through mask **30**, thereby increasing the accuracy to which object **22** can be printed.

(28) Controller **36** may be communicatively coupled to mask **30**, beam scanner **26**, light source **24** and height adjustor **39** via control signal paths **38a**, **38b**, **38c** and **38d**, respectively (e.g., electrical signal paths). Controller **36** may control the addressable pixels of mask **30** such that the transparent pixels of mask **30** correspond to a cross section of an object to be printed (e.g., a layer of that object). Controller **36** may control beam scanner **26** to selectively scan a light beam across regions of mask **30** with transparent pixels, while substantially avoiding regions of mask **30** with only opaque pixels. Often times, the transparent pixels only account for a portion of the total pixels (e.g., 30%, 50%, etc.). Assuming those transparent pixels are aggregated in certain regions (which is often the case), only those regions of the mask are scanned, which substantially reduces the number of opaque pixels that are irradiated unnecessarily, in turn reducing the heating of mask **30** and resin **18**. Specific examples of the scanning of light beam **28** will be provided below.

(29) Controller **36** may also control light source **24**. For instance, to further reduce the heating of mask **30**, controller **36** may turn off light source **24** while light beam **28** is being repositioned by scanner **26** from one region of mask **30** with transparent pixels to another region of mask **30** with transparent pixels. Controller **36** may also control height adjustor **39** to control the vertical position of height extractor **20**, and consequently of object **22** that is affixed to height extractor **20**.

(30) As depicted in FIG. 2, light source **24** may include laser source **40** that generates laser beam **42**, and a beam expander **44** which transforms the collimated and focused laser beam **42** into a collimated and defocused light beam **28**. For the sake of conciseness, collimated and defocused light beam **28** is simply referred to as “light beam” **28** throughout the description. As depicted in FIG. 2, a diameter, d_2 , of the cross section of light beam **28** may be larger than a diameter, d_1 , of the cross section of laser beam **42**.

(31) FIG. 3A depicts mask **30** during an exposure time duration, during which time, some pixels of mask **30** are controlled to be in a transparent state while other pixels of mask **30** are controlled to be in an opaque state (although mask **30** is not depicted at a level of detail in which the individual pixels are visible). For clarity of illustration, region **50** of mask **30** with opaque pixels is depicted with a gray shading, while region **52** of mask **30** with transparent pixels is depicted in white (i.e., without any shading). It is understood that a light beam scanning across mask **30** will pass through region **52** of the mask (and cure portions of layer **34** of liquid resin **18**), while the light beam will not pass through region **50** of the mask. The shape of region **52** is chosen to approximately correspond to a cross section **53** of an object that is to be printed (see cross section **53** depicted as an inset in FIG. 3A). Typical dimensions of mask **30** (i.e., in the diagonal direction) may measure 13.3 inches, while it is contemplated that the dimensions of mask **30** will increase in the future, allowing for the printing of objects with larger dimensions.

(32) FIG. 3B depicts a magnified view of portion **54** of the mask **30** depicted in FIG. 3A, in which individual pixels (e.g., electrically modulated liquid crystal pixel elements) are visible in the magnified view. Reference numeral **56** labels one of the opaque pixels, while reference numeral **58** labels one of the transparent pixels of mask **30**. For clarity of illustration, opaque pixels are depicted in gray shading, while clear pixels are depicted in white (i.e., without any shading). It is understood that the visualization of pixels in FIG. 3B is merely a schematic illustration, and may not depict an actual representation thereof. For instance, pixels are depicted with square boundaries in FIG. 3B, but other boundary shapes are possible, such as a rectangular boundary, an oval boundary, a circular boundary, etc. The physical construction of a pixel (e.g., liquid crystal sandwiched between two electrodes) is well known in the art, and will not be discussed herein for the sake of conciseness.

(33) FIG. 4A depicts cross section **60** of light beam **28** at the surface of mask **30**. For conciseness of discussion, cross section **60** may be referred to as a “beam spot,” but if the “illuminated” area of mask **30** comprises transparent pixels, it is understood that the “beam spot” may not actually be

visible, as light beam **28** may shine through mask **30** without reflecting off of the surface of mask **30**.

(34) FIG. **4B** depicts a magnified version of portion **54** of the mask **30** depicted in FIG. **3B**. As shown in FIG. **4B**, the diameter, d_2 , of beam spot **60** may be an order of magnitude (or more) greater than the cross sectional dimension, w , of each of the respective pixels. In one embodiment of the invention, the diameter, d_2 , is at least ten times the cross-sectional dimension, w , of each of the respective pixels. In another embodiment of the invention, the diameter, d_2 , is at least one hundred times the cross-sectional dimension, w , of each of the respective pixels. As an example, w may measure 25-150 μm , whereas d_2 may measure 10 mm. In another embodiment of the invention, the diameter, d_2 , of beam spot **60** may be dynamically adjusted based on the cross-sectional dimensions of the object to be fabricated. If the cross-sectional dimensions of the object to be fabricated are on the order of centimeters, d_2 may measure 1 centimeter. If the cross-sectional dimensions of the object to be fabricated are on the order of millimeters, d_2 may measure 1 millimeter. Such dynamical adjustment of the beam spot diameter may further reduce the illumination of opaque pixels (and consequently reduce the heating of the liquid resin), while preserving the throughput for objects having larger cross-sectional dimensions.

(35) FIG. **5** depicts beam path **62** of a light beam performing a raster scan of a transparent region **52** of mask **30**. Beam spot **60** continuously travels (i.e., sweeps) across the surface of mask **30** along beam path **62**. The beam path **62** may be determined by controller **36** based on the locations of the transparent pixels in mask **30** (i.e., beam path **62** is chosen to illuminate the transparent pixels of mask **30** in a uniform manner). It is understood that a thin border of opaque pixels surrounding transparent region **52** may also be illuminated, in order to allow for the possibility for some inaccuracy in the control of the location of beam spot **60** on mask **30**, and also allowing for the possibility for some inaccuracy in the control of the beam spot diameter. However, the number of opaque pixels (e.g., in the thin border) that are illuminated may be minimized to minimize the heating of mask **30** by light beam **28**. In a scenario where transparent pixels are concentrated in a single region (such as in the example of FIG. **5**), it is possible that no more than 1% of the opaque pixels are illuminated by light beam **28** (during the printing of a single cross section of object **22**). In a scenario where transparent pixels are concentrated in multiple regions (such as in the example of FIG. **8**), it is possible that no more than 10% of the opaque pixels are illuminated (during the printing of a single cross section of object **22**). It is understood that the beam spot of successive “rows” of the raster scan may overlap by a few pixels so as to allow for region **52** to be scanned with uniform light intensity (i.e., uniform intensity, as averaged out over time). Further, it is understood that the beam path **62** depicted in FIG. **5** may be traced out several times by light beam **28** (one time in the direction depicted in FIG. **5**; the next time, following the path in the reverse direction; and then in the next time, following the direction depicted in FIG. **5**, and so on). Repeatedly performing a fast scan of a region (e.g., performing **10** quick traversals through beam path **62**) may be more optimal than performing a single slow scan of a region (e.g., perform a single traversal through beam path **62**), as the heating of resin **18** may be spread out more uniformly across layer **34**.

(36) FIG. **6** depicts opaque regions **50** and transparent regions **52** of mask **30**, during another exposure time duration. In FIG. **6**, the transparent regions **52** are arranged within a “thin strip” with width dimensions less than the diameter, d_2 , of the beam spot **60**. Consequently, as shown in FIG. **7**, light beam **28** may be repeatedly scanned in a “back and forth” manner along beam path **62** so as to illuminate transparent regions **52** of mask **30**. During such scanning, it is understood that some opaque pixels in opaque region **50** may also be scanned by light beam **28** (i.e., when light beam **28** passes from one transparent region to another), but number of opaque pixels that are scanned is minimized to a large degree, as compared to the scenario in which the entire mask were raster scanned.

(37) FIG. **8** depicts opaque regions **50** and transparent regions **52** of mask **30**, during another

exposure time duration. In the example of FIG. 8, region **64a** of mask **30** includes a high concentration of transparent pixels, likewise for region **64c**, and region **64a** is separated from region **64c** by region **64b** with only opaque pixels. FIG. 9 depicts the beam paths **62a**, **62b** that may be followed by light beam **28** to scan the transparent pixels of mask **30**. In one scenario, light beam **28** may scan the transparent pixels within region **64a** by repeatedly following beam path **62a**. Beam scanner **26** may then reposition light beam **28** to region **64c** without light beam **28** scanning region **64b** with only opaque pixels. During the repositioning of light beam **28**, controller **36** may turn off light source **24** or control a blocking element (not depicted) to block light beam **28**. For instance, the blocking element may include a shutter of light source **24** that can be controlled by controller **36** to block light beam **28**. After the repositioning, light beam **28** may scan the transparent pixels within region **64c** by repeatedly following beam path **62b**. It is noted that the scanning speed of laser beam **28** within region **64a** may differ from the scanning speed of laser beam **28** within region **64c**. For instance, smaller regions may be scanned at a slower speed than larger regions.

(38) FIG. 10 depicts a scanning scheme that minimizes the scanning of opaque pixels without the need to turn off light source **24** or block light beam **28**. In the scanning scheme of FIG. 10, light beam **28** similarly scans the transparent pixels within region **64a** by repeatedly following beam path **62a**. However, during the repositioning of the light beam **28** from region **64a** to **64c**, light beam **28** scans along beam path **62c**. Beam path **62c** may be a shortest path through region **64b** that connects beam path **62a** within region **64a** and beam path **62b** within region **64c**. After the repositioning, light beam **28** may scan the transparent pixels within region **64c** by repeatedly following beam path **62b**.

(39) FIG. 11 depicts a 3D printing system **101** that employs multiple light beams (e.g., two light beams) for scanning mask **30**. Beam scanner **26a** may scan light beam **28a** from light source **24a** selectively across certain regions of mask **30**, and depending on whether the scanned pixels are transparent or opaque, filtered light beam **32a** may be transmitted through mask **30** and cure a portion of resin in layer **34**. Similarly, beam scanner **26b** may scan light beam **28b** from light source **24b** selectively across other regions of mask **30**, and depending on whether the scanned pixels are transparent or opaque, filtered light beam **32b** may be transmitted through mask **30** and cure a portion of resin in layer **34**. As an example, light beam **28a** could follow beam path **62a** in FIG. 10, and light beam **28b** could follow beam path **62b** in FIG. 10. While increasing the cost of 3D printing system **101**, multiple light beams may provide for a faster throughput (i.e., printing speed) as compared a 3D printing system that employs a single light beam. For ease of depiction, controller **36** has not been illustrated in FIG. 11, but it should be apparent that controller **36** may be used to control beam scanner **26a** and **26b** and other previously described components of FIG. 11.

(40) FIG. 12A depicts flow chart **102** of a method to print a cross section of a three-dimensional object with reduced heat generation. At step **104**, controller **36** may control, during an exposure time duration, a first subset of the pixels to be transparent at locations corresponding to the cross section of a (to be printed) three-dimensional object, and a second subset of the pixels to be opaque at locations not corresponding to the cross section of the three-dimensional object. At step **106**, controller **36** may control beam scanner **26**, during the same exposure time duration as step **104**, to scan light beam **28** across at least one region of the mask having at least some pixels that are controlled to be transparent. The scanning may be performed such that light beam **28** is always incident on at least one of the pixels of mask **30** that is controlled to be transparent during the printing of the cross section of the three-dimensional object. Such a scanning scheme was illustrated in FIGS. 5, 7 and 9. The heat reduction, of course, is most pronounced when the transparent pixels only account for a small (or smaller) portion of the total number of pixels (e.g., less than 30%-50% of the total number of pixels).

(41) FIG. 12B depicts flow chart **108** of another method to print a cross section of a three-dimensional object with reduced heat generation. At step **110**, controller **36** may control, during an

exposure time duration, a first subset of the pixels to be transparent at locations corresponding to the cross section of a (to be printed) three-dimensional object, and a second subset of the pixels to be opaque at locations not corresponding to the cross section of the three-dimensional object. At step **112**, controller **36** may control beam scanner **26**, during the same exposure time duration as step **110**, to scan light beam **28** across at least one region of the mask having at least some pixels that are controlled to be transparent. The scanning may be performed such that at most ten percent of the pixels that are controlled to be opaque are scanned by light beam **28** during the printing of the cross section of the three-dimensional object. Such a scanning scheme was illustrated in FIGS. **5**, **7**, **9** and **10**. Again, the heat reduction is most pronounced when the transparent pixels only account for a small (or smaller) portion of the total number of pixels (e.g., less than 30%-50% of the total number of pixels).

(42) FIG. **13A** depicts flow chart **200** of a method to scan light beam **28** across a surface of a mask of a 3D printing system. At step **202**, light beam **28** may be repeatedly scanned across a first region of mask **30** that includes at least some transparent pixels. Step **202** was described above by the scanning of transparent pixels within region **64a** in FIG. **9**. At step **204**, light beam **28** may be repositioned from the first region to a third region of the mask that includes at least some transparent pixels, without scanning a second region of the mask that separates the first region from the third region, the second region of the mask including only opaque pixels. Step **204** was described above in FIG. **9** in the repositioning of light beam **28** from region **64a** to region **64c**. At step **206**, light beam **28** may be repeatedly scanned across the third region of the mask that includes at least some transparent pixels. Step **206** was described above by the scanning of transparent pixels within region **64c** in FIG. **9**.

(43) FIG. **13B** depicts flow chart **208** of a method to scan light beam **28** across a surface of a mask of a 3D printing system. At step **210**, light beam **28** may be repeatedly scanned across a first region of mask **30** that includes at least some transparent pixels. Step **210** was described above by the scanning of transparent pixels within region **64a** in FIG. **10**. At step **212**, light beam **28** may be scanned along a path within a second region that separates the first region from a third region, the second region including only opaque pixels, and the third region including at least some transparent pixels, the path being a shortest path that connects a beam path in the first region and a beam path in the third region. Step **212** was described above in FIG. **10** by the scanning of light beam **28** along beam path **62c**. At step **214**, light beam **28** may be repeatedly scanned across the third region of the mask that includes at least some transparent pixels. Step **214** was described above by the scanning of transparent pixels within region **64c** in FIG. **10**.

(44) In practice, there may be some non-uniformity in the light transmissivity across respective pixels of the mask (e.g., more than 10% variation across the pixels). For example, even if a pixel is controlled to be (fully) transparent, it may only be 95% transparent to light due to defects, aging of the pixel, etc. Therefore, for the sake of clarity, it is noted that the above-mentioned “transparent pixel” may refer to a pixel that is 100% transparent to light, 99% transparent to light, 95% transparent to light, etc. Likewise, the above-mentioned “opaque pixel” may refer to a pixel that is 100% opaque to light, 99% opaque to light, 95% opaque to light, etc.

(45) FIG. **14** depicts flow chart **250** of a method for printing a cross section of a three dimensional object that includes scanning light beam **28** across a surface of a mask of a 3D printing system in such a way that the scanning compensates for the non-uniformity in the light transmissivity across respective pixels of the mask. At step **252**, a uniformity in a light transmissivity across respective pixels of the mask may be assessed. Such an assessment may comprise controlling all pixels to be (fully) transparent, illuminating the entire mask (e.g., scanning a light beam across the entire mask), and measuring the intensity of the light transmitted by each of the pixels. During this initial assessment, it is assumed that the light intensity of the light beam itself is fairly uniform, regardless of whether the light beam is shining near the central region or the peripheral regions of the mask. The respective locations of any pixels with a less-than-expected light intensity may be identified

(e.g., an attenuated light intensity relative to other pixel elements).

(46) At step **254**, controller **36** may control, during an exposure time duration, a first subset of the pixels to be transparent at locations corresponding to the cross section of a (to be printed) three-dimensional object, and a second subset of the pixels to be opaque at locations not corresponding to the cross section of the three-dimensional object. At step **256**, controller **36** may control beam scanner **26**, during the same exposure time duration as step **104**, to scan light beam **28** across at least one region of the mask having at least some pixels that are controlled to be transparent. The scanning may be performed in a manner that compensates for the non-uniformity in the light transmission across respective pixels in the at least one region of the mask. The compensation may include: (i) varying a light intensity of the light beam while the light beam is scanned over the at least one region, (ii) varying a scan speed of the light beam while the light beam is scanned over the at least one region, or (iii) varying a number of times the light beam is repeatedly scanned over the at least one region. More specifically, for those regions where the pixels are known (via the assessment in step **252**) to output an attenuated light output, the light intensity of the light beam may be increased, the scanning speed of the light beam may be decreased and/or the number of scanning passes through those regions may be increased so as to compensate for the attenuated light output.

(47) As is apparent from the foregoing discussion, aspects of the present invention involve the use of various computer systems and computer readable storage media having computer-readable instructions stored thereon. FIG. **15** provides an example of system **300** that may be representative of any of the computing systems (e.g., controller **36**) discussed herein. Note, not all of the various computer systems have all of the features of system **300**. For example, certain ones of the computer systems discussed above may not include a display inasmuch as the display function may be provided by a client computer communicatively coupled to the computer system or a display function may be unnecessary. Such details are not critical to the present invention.

(48) System **300** includes a bus **302** or other communication mechanism for communicating information, and a processor **304** coupled with the bus **302** for processing information. Computer system **300** also includes a main memory **306**, such as a random access memory (RAM) or other dynamic storage device, coupled to the bus **302** for storing information and instructions to be executed by processor **304**. Main memory **306** also may be used for storing temporary variables or other intermediate information during execution of instructions to be executed by processor **304**. Computer system **300** further includes a read only memory (ROM) **308** or other static storage device coupled to the bus **302** for storing static information and instructions for the processor **304**. A storage device **310**, for example a hard disk, flash memory-based storage medium, or other storage medium from which processor **304** can read, is provided and coupled to the bus **302** for storing information and instructions (e.g., operating systems, applications programs and the like).

(49) Computer system **300** may be coupled via the bus **302** to a display **312**, such as a flat panel display, for displaying information to a computer user. An input device **314**, such as a keyboard including alphanumeric and other keys, may be coupled to the bus **302** for communicating information and command selections to the processor **304**. Another type of user input device is cursor control device **316**, such as a mouse, a trackpad, or similar input device for communicating direction information and command selections to processor **304** and for controlling cursor movement on the display **312**. Other user interface devices, such as microphones, speakers, etc. are not shown in detail but may be involved with the receipt of user input and/or presentation of output.

(50) The processes referred to herein may be implemented by processor **304** executing appropriate sequences of computer-readable instructions contained in main memory **306**. Such instructions may be read into main memory **306** from another computer-readable medium, such as storage device **310**, and execution of the sequences of instructions contained in the main memory **306** causes the processor **304** to perform the associated actions. In alternative embodiments, hard-wired circuitry

or firmware-controlled processing units may be used in place of or in combination with processor **304** and its associated computer software instructions to implement the invention. The computer-readable instructions may be rendered in any computer language.

(51) In general, all of the above process descriptions are meant to encompass any series of logical steps performed in a sequence to accomplish a given purpose, which is the hallmark of any computer-executable application. Unless specifically stated otherwise, it should be appreciated that throughout the description of the present invention, use of terms such as “processing”, “computing”, “calculating”, “determining”, “displaying”, “receiving”, “transmitting” or the like, refer to the action and processes of an appropriately programmed computer system, such as computer system **300** or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within its registers and memories into other data similarly represented as physical quantities within its memories or registers or other such information storage, transmission or display devices.

(52) Computer system **300** also includes a communication interface **318** coupled to the bus **302**. Communication interface **318** may provide a two-way data communication channel with a computer network, which provides connectivity to and among the various computer systems discussed above. For example, communication interface **318** may be a local area network (LAN) card to provide a data communication connection to a compatible LAN, which itself is communicatively coupled to the Internet through one or more Internet service provider networks. The precise details of such communication paths are not critical to the present invention. What is important is that computer system **300** can send and receive messages and data through the communication interface **318** and in that way communicate with hosts accessible via the Internet.

(53) Thus, methods and systems for photocuring liquid resin with reduced heat generation has been described. It is to be understood that the above-description is intended to be illustrative, and not restrictive. Many other embodiments will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled.

Claims

1. A vat polymerization printer, comprising: a tank configured for containing a photo-curable liquid resin; a light source configured to emit a light beam according to inputs from a controller; a mask having pixels configurable to be individually transparent or opaque to portions of the light beam, wherein a diameter of a cross section of the light beam is greater than a cross-sectional dimension of each of the respective pixels; a beam scanner configured to scan the light beam across the mask according to inputs from the controller; and the controller comprising a memory and a processor, the memory storing instructions that, when executed, cause the processor to control the vat polymerization printer to print a cross section of a three-dimensional object by: the controller configured to control, during an exposure time duration, a first subset of the pixels of the mask to be transparent at locations corresponding to the cross section of the three-dimensional object, and a second subset of the pixels of the mask to be opaque at locations not corresponding to the cross section of the three-dimensional object, wherein, during the exposure time duration, and as a result of the control of the first and second subset of the pixels, (i) a first region of the mask includes at least some pixels that are controlled to be transparent, the first region of the mask being surrounded by a first border with pixels that are controlled to be opaque, (ii) a second region of the mask includes only pixels that are controlled to be opaque, and (iii) a third region of the mask includes at least some pixels that are controlled to be transparent; and the controller configured to control, during the exposure time duration, the light source and the beam scanner to: illuminate the mask with the light beam such that at most ten percent of the pixels that are controlled to be opaque are scanned by the light beam during the printing of the cross section of the three-dimensional object,

repeatedly scan the light beam across the first region of the mask, illuminate the first border, and turn off the light source while the beam scanner repositions the light beam between the first region of the mask that includes at least some pixels that are controlled to be transparent to the third region of the mask that includes at least some pixels that are controlled to be transparent, the third region of the mask being separated from the first region of the mask by the second region of the mask that includes only pixels that are controlled to be opaque.

2. The vat polymerization printer of claim 1, wherein the diameter of the cross section of the light beam is at least ten times the cross-sectional dimension of each of the respective pixels.
 3. The vat polymerization printer of claim 1, wherein the diameter of the cross section of the light beam is at least a hundred times the cross-sectional dimension of each of the respective pixels.
 4. The vat polymerization printer of claim 1, wherein the light source comprises: a laser source configured to emit a laser beam; and a beam expander configured to generate the light beam from the laser beam, wherein the diameter of the cross section of the light beam is greater than a diameter of a cross section of the laser beam.
 5. The vat polymerization printer of claim 1, wherein the instructions further cause the processor to determine a scan path for the light beam based on respective locations of the pixels that are controlled to be transparent during the exposure time duration.
 6. The vat polymerization printer of claim 1, wherein the pixels comprise electrically modulated liquid crystal pixel elements.
 7. The vat polymerization printer of claim 1, further comprising: an extraction plate disposed within the tank to which the three-dimensional object, formed from cured portions of the photo-curing liquid resin, is affixed; and a height adjustor configured to control a vertical position of the extraction plate above the mask.
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